

Toward single-shot characterization of nonlinear optical refraction, absorption, and scattering of turbid media

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(Received 26 June 2020; accepted 31 July 2020; published 1 September 2020)

Disordered media are often susceptible to optical damage even at low irradiation doses, since the corresponding microscopic structure may suffer from processes such as bleaching, chemical reactions, or morphological changes. The disorder often introduces linear and nonlinear (NL) light scattering, and currently it is a challenge to perform single-shot characterization of NL refraction, NL absorption, and NL scattering using the more common NL transmittance techniques. This work discusses how the scattered light imaging method and D4 σ techniques can be combined to characterize the NL optical constants of thick turbid media. Therefore, it should help researchers aiming to draw a detailed physical picture of turbid samples at the single laser shot level.

DOI: [10.1103/PhysRevA.102.033503](https://doi.org/10.1103/PhysRevA.102.033503)

I. INTRODUCTION

In several relevant systems, disorder is a prominent feature, such as the white color of bones and some insects, and the iridescent colors of plants and animals [1]. The rich physics within disordered systems can be exploited in optics to produce random lasers [2,3], look around walls [4], or focus light in time and space through turbid media [5]. They also can be used as a testbed for several concepts in condensed-matter systems. For example, they may exhibit Anderson localization [6], spin-glass and turbulencelike behaviors [7], Levy flights and replica symmetry breaking [8] or a Floquet phase [9], and discrete-time crystals [10]. The nonlinear (NL) response of such systems is very important in several of these applications, such that the characterization and understanding of the underlying microscopic behavior has a strong significance.

Among the possible NL optical responses, the third-order susceptibility $\chi^{(3)}$ is often relevant for being present irrespective of the underlying microscopic symmetries [11,12]. The value of $\chi^{(3)}$ in bulk samples is often determined through NL refraction and NL extinction measurements performed through NL transmittance experiments, such as the Z-scan [13] or D4 σ [14,15] methods. When NL scattering is negligible, the NL extinction gives pure information about NL absorption. However, while NL transmittance techniques are appropriate for homogeneous materials, they may face difficulties to completely characterize samples exhibiting a considerable amount of scattering [16,17]. To ensure that some proposed power-loss contribution is due to NL scattering, it is necessary to evaluate the NL scattering off of the incident beam's optical axis [18] or to vary the optical aperture of the detection system [17,19]. Another problem is that the standard Z-scan NL refraction signal-to-noise ratio can degrade signifi-

cantly when probing scattering media, other approaches being more suited [20,21]. This is an important limitation to the standard Z-scan characterization, since biomaterials [22] and colloids, in general, often contain scattering centers. Another intrinsic difficulty with the traditional Z-scan for scattering samples is that several samples of interest are sensitive to optical damage [19,21], such that NL spectroscopy techniques capable of single-shot measurements of optical constants are particularly desirable.

The scattered light imaging method (SLIM) [20,23] uses a camera to take a snapshot of the transversely scattered light along the beam propagation axis in a thick sample. The information contained in each snapshot can be used to characterize laser beam parameters, such as the beam waist and the divergence angle, for sufficient long samples [23]. It was also previously shown that the data obtained are enough to determine the NL refraction coefficient with high sensitivity in a sample containing scattering inclusions [20]. Another important feature of this technique is that for sufficiently low optical pulse rates, the analysis can be performed on snapshots containing data from a single optical pulse. Therefore, the SLIM seems well suited to the challenge of NL spectroscopy of scattering samples susceptible to optical damage. The SLIM is also competitive from a cost and complexity perspective since it requires just a simple imaging system. This contrasts with the spectral reshaping technique [21], which can perform sensitive NL spectroscopy measurements of scattering samples, but requires amplified femtosecond optical pulses as an input to a spectral pulse shaper. While SLIM can be used to characterize the transverse scattering at a single shot, a camera along the optical axis, as in the D4 σ method [14,15], can complement these measurements with data from forward scattering and power extinction using a single laser shot. The D4 σ method positions a camera after the NL sample to collect the transverse profile of the beam transmitted by the NL sample. Contrarily to the usual Z-scan method, which uses a small aperture to determine the

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NL phase variations, the $D4\sigma$ collects the whole transmitted beam and determines the NL phase variations through the transverse irradiance moments. The first and second moments (beam centroid and width) are sufficient to measure NL refraction [15] and can be determined from a single snapshot. A positive side effect of using the entire beam profile is that the beam pointing instability noise is virtually eliminated [14], thus enhancing the signal-to-noise ratio and sensitivity. The NL extinction for the transmitted beam can also be evaluated with the determination of the optical power reaching the camera. Thus, the $D4\sigma$ method is particularly relevant for the present work since it allows one to characterize the NL extinction and the NL forward-scattering angles either from the beam divergence directly [15] or using the Fourier transform property of a lens [14].

Even though the SLIM is simple from an experimental point of view, its analytical description is more difficult than those from the more usual NL transmittance techniques. In the latter, the thin sample approximation is often used, where diffraction is ignored within the NL material [13,15]. However, the thin sample approximation cannot be used in the SLIM. A thick NL sample is required to understand how the NL response at the beam waist affects the beam divergence [20]. The signals extracted in SLIM are based on the detailed evolution of the beam parameters within a thick NL scattering medium.

The present work discusses how to describe the propagation of optical beams within thick turbid samples subject to NL absorption and elastic NL scattering. Initially, some microscopic aspects are considered. Later, exact results are shown for the beam power evolution and the beam divergence in terms of effective beam parameters. The theoretical results shown below can be directly related to SLIM and $D4\sigma$ experimental parameters. Later, the effects of NL light scattering towards the transverse and forward detectors are considered and how it affects the measurements. The analysis is performed initially for cw beams, and later extended for Gaussian optical pulses. Finally, a discussion is presented of how several measurable parameters modify the experimental SLIM signals.

II. BEAM PROPAGATION IN A THICK TURBID NONLINEAR MEDIUM

A. Some microscopic aspects of elastic scattering

In the linear optics regime, the Kramers-Krönig relations ensure that a frequency-dependent refractive index is associated with a respective absorption profile. Thus, refraction and absorption are present in all materials. Scattering requires some inhomogeneity within the material, and it depends on the detailed morphology, composition, and distribution of these structures [24]. While scattering is not present in several effective models, it may be an important feature of real samples and unavoidable in some fabrication processes.

Since scattering (either linear or NL) removes power from the pump beam, the power extinction cross section, C_{ext} , can be expressed as

$$C_{\text{ext}} = C_{\text{abs}} + C_{\text{scatt}}, \quad (1)$$

where C_{abs} and C_{scatt} are the absorption and scattering cross sections, respectively. For simplicity, consider initially a single small sphere of radius a and dielectric constant ϵ_1 embedded in a medium whose dielectric constant is ϵ_0 and refractive index n_0 . Then, it can be stated in the quasistatic approximation that [24]

$$C_{\text{abs}} = 4\pi k a^3 \text{Im} \left\{ \frac{\epsilon_1 - \epsilon_m}{\epsilon_1 + 2\epsilon_m} \right\}, \quad (2)$$

$$C_{\text{scatt}} = \frac{8}{3} \pi k^4 a^6 \left| \frac{\epsilon_1 - \epsilon_m}{\epsilon_1 + 2\epsilon_m} \right|^2, \quad (3)$$

where $k = 2\pi n_0/\lambda$ is the wave number of light with a free-space wavelength λ . C_{abs} is proportional to the inclusion volume $a^3 \sim V$, while C_{scatt} is proportional to $a^6 \sim V^2$. Therefore, for small particles where $ka \ll 1$, it is often valid that the power extinction due to scattering is negligible and only the absorption should be considered. However, the scattering contribution may be a significant extinction mechanism for transparent materials since $C_{\text{abs}} \approx 0$, but the dielectric constant contrast $|\epsilon_1 - \epsilon_m|$ can be large. Another situation where scattering should be more relevant is close to resonances. For example, in metal colloids, the condition $\text{Re}(\epsilon_1 + 2\epsilon_m) = 0$ is satisfied at the plasmon resonance and it diverges linearly in C_{abs} and quadratically in C_{ext} . Thus, significant linear scattering can be expected under strong spatial variations in the dielectric constant or close to structural resonances. As practical examples, strong elastic NL scattering was observed in colloids containing several morphologies of metallic nanostructures [18,19].

The above reasoning can be extended to the NL regime in a dilute turbid medium. The Maxwell-Garnett model can be used to obtain an effective medium theory with local field corrections [25]. However, it should be remarked that a negligible scattering in the linear optics regime does not translate directly to the domain of NL optics. A transparent colloid whose solution and inclusions are index matched in the linear regime may exhibit a strong NL scattering [26]. Therefore, the effective medium extinction coefficient may require corrections due to scattering.

In the Rayleigh-Gans approximation [24,26], the extinction coefficient due to linear and NL scattering is written as

$$\alpha_s = g_s (n_1 - n_0)^2, \quad (4)$$

where n_1 is the total refractive index of the inclusion and g_s is a constant that depends on the wavelength, concentration, size, and shape of the inclusion. Since in the present work we are interested in elastic scattering, it is assumed that the refractive indices can be expanded as functions of irradiance as $n_i = n_{i,0} + n_{i,2}I$. From Eq. (4), it should be more explicit that an index matching in the linear regime may significantly change in the NL regime, particularly because intense light beams may substantially modify the molecular environment close to the inclusion surface [26].

B. Beam power propagation inside the sample

Consider the wave equation for a linearly polarized monochromatic electromagnetic wave propagating along the

z axis in the paraxial regime with electric field given by $\mathbf{E} = \frac{1}{2}\epsilon\mathcal{E}(\mathbf{r}, z)e^{i(kz - \omega t)} + \text{c.c.}$, where ω is the angular frequency, $k = \frac{\omega}{c}n_0$ is the wave number, n_0 is the effective linear refractive index inside the material, ϵ is the polarization vector, and $\mathbf{r}$ is a vector transverse to z . The beam temporal profile can also play an important role in the description of nonlinear optical processes with pulsed lasers. For simplicity, the present discussion is limited to materials with an instantaneous response and with negligible dispersion.

The beam envelope $\mathcal{E}(\mathbf{r}, z)$ propagates along z according to

$$\left[\nabla_T^2 + 2ik\frac{\partial}{\partial z} + 2k_0^2n_0n' + ik\alpha\right]\mathcal{E}(\mathbf{r}, z) = 0, \quad (5)$$

where $n' = n'[I(\mathbf{r}, z)]$ and $\alpha = \alpha[I(\mathbf{r}, z)]$ represent the medium full NL refraction and extinction coefficients, respectively, as a function of the irradiance $I(\mathbf{r}, z)$, $k_0 = k/n_0$, and ∇_T^2 is the Laplacian along the beam coordinates transverse to z . In our convention, α is real and can represent both absorption and scattering processes that attenuate the beam under propagation. The integral of the irradiance along the plane transverse to the propagation direction gives the optical power $P(z)$ at a plane z ,

$$P(z) = \int d^2\mathbf{r}I(\mathbf{r}, z) \quad (6)$$

$$= \frac{1}{2}c\epsilon_0n_0 \int d^2\mathbf{r}|\mathcal{E}(\mathbf{r}, z)|^2, \quad (7)$$

where c is the vacuum light speed, ϵ_0 is the vacuum electrical permittivity, and $d^2\mathbf{r} = dxdy$ indicates that the integral is being considered along the whole transverse plane.

For a beam that propagates inside a thick NL medium, it is often considered that Eq. (5) must be calculated to obtain the evolution of $P(z)$ through (7). However, notice that z and $\mathbf{r}$ are orthogonal coordinates such that we can obtain a differential equation for the evolution of $P(z)$ using (5). More clearly, the operator $\frac{\partial}{\partial z}$ can act directly upon the integrand $|\mathcal{E}(\mathbf{r}, z)|^2$ in Eq. (7). For convenience, we can rewrite Eq. (5) as

$$\frac{\partial \mathcal{E}}{\partial z} = \left[\frac{i}{2k}\nabla_T^2 + ik_0n' - \frac{\alpha}{2}\right]\mathcal{E}, \quad (8)$$

such that

$$\frac{dP}{dz} = \frac{1}{2}c\epsilon_0n_0 \int d^2\mathbf{r}\left(\mathcal{E}^*\frac{\partial \mathcal{E}}{\partial z} + \mathcal{E}\frac{\partial \mathcal{E}^*}{\partial z}\right), \quad (9)$$

where the arguments of the optical power, the field, and its complex conjugate are implicitly assumed from now on in larger expressions for improved clarity. The integrand in (9) can be expressed with the help of (8) as

$$\mathcal{E}^*\frac{\partial \mathcal{E}}{\partial z} + \mathcal{E}\frac{\partial \mathcal{E}^*}{\partial z} = \frac{1}{k}\text{Im}[\mathcal{E}^*\nabla_T^2\mathcal{E}] - \alpha|\mathcal{E}|^2. \quad (10)$$

As a remark, Eq. (10) holds even when NL refraction is considered. Also, notice that since (10) appears under the integral sign and we are interested in fields with a finite transverse profile, $\mathcal{E}$ and all of its transverse derivatives become zero when $|\mathbf{r}| \rightarrow \infty$. Therefore, the terms involving transverse derivatives in (10) cancel under the integral sign after two integration by parts in (9) and $\int d^2\mathbf{r}\mathcal{E}^*\nabla_T^2\mathcal{E} = \int d^2\mathbf{r}\mathcal{E}\nabla_T^2\mathcal{E}^*$,

such that

$$\frac{dP}{dz} = - \int d^2\mathbf{r}\alpha I. \quad (11)$$

Equation (11) is an exact result and is valid for arbitrary paraxial propagation problems, involving either NL refraction or NL extinction. Equation (11) is not yet very practical as it stands because of the transverse integral. However, to demonstrate how to apply Eq. (11) for beam propagation, it is assumed for definiteness and simplicity that

$$\alpha = \alpha_1 + \alpha_2 I, \quad (12)$$

where α_1 indicates the linear extinction and α_2 is the two-photon extinction coefficient. These extinction coefficients consider both absorption and scattering. Then, Eq. (11) becomes

$$\frac{dP}{dz} = -\alpha_1 P - \alpha_2 \int d^2\mathbf{r}I^2. \quad (13)$$

While it seems at first that Eq. (13) is not very useful since it involves both the irradiance and the beam power, it is possible to define the inverse beam area parameter as

$$\gamma(z) = \frac{\int d^2\mathbf{r}I^2(\mathbf{r}, z)}{[\int d^2\mathbf{r}I(\mathbf{r}, z)]^2} = \frac{\int d^2\mathbf{r}I^2(\mathbf{r}, z)}{P^2(z)}, \quad (14)$$

where $\gamma(z)$ has units of $(\text{Area})^{-1}$, and it is equivalent to the effective beam area commonly used in optical fibers [11], but here is meant to describe diffracting beams. For example, if the beam transverse profile is Gaussian, then $I(\mathbf{r}, z) \propto e^{-2\mathbf{r}^2/w^2(z)}$ and $\gamma(z) = [\pi w^2(z)]^{-1}$, where $w(z)$ is the beam width at z . In terms of γ , Eq. (11) becomes exactly

$$\frac{dP}{dz} = -\alpha_1 P - \alpha_2 \gamma P^2. \quad (15)$$

Equation (15) is a significant improvement to Eq. (11) since it involves only two functions of z , P , and γ . Therefore, to understand the optical power evolution inside a NL optical medium, it is possible to exchange the full solution of Eq. (5) by this first-order differential equation. There is one possible caveat here: γ also evolves along z , and it can be difficult to obtain its z dependence. This is not a major problem in experiments since the beam transverse profile can be measured using SLIM and γ can be directly determined from the experimental data. Therefore, it is assumed that γ is a known function of z , such that Eq. (15) can be solved exactly. Given a known γ , it can be shown (see Appendix A) that the optical power evolves inside the sample according to

$$P(z) = \frac{P_0 e^{-\alpha_1 z}}{1 + \alpha_2 P_0 \int_0^z \gamma(z') e^{-\alpha_1 z'} dz'}, \quad (16)$$

where z represents the distance that the light beam has propagated inside the NL sample, and P_0 is the incident beam power. Equation (16) is a relevant result and worth noting, given that it becomes identical to the best result for thick media inferred in the detailed heuristic analysis by Hermann [27] when it is considered that $\gamma(z) = [\pi w^2(z)]^{-1}$. In principle, $P(z)$ can be determined both from SLIM and the D4 σ methods. However, NL scattering modifies the SLIM measurement, as will be discussed below. On the other hand, a NL transmittance

technique is sensitive to the total beam power extinction and may fix the total extinction coefficient α_2 . Therefore, both the NL transmission and NL scattering measurements can complement each other to determine the NL absorption and scattering coefficients.

As a final comment, it will be useful in the next section to define an effective intensity at a z plane through

$$\bar{I}(z) = \gamma(z)P(z). \quad (17)$$

C. Effective beam divergence along the sample

To determine the NL refraction coefficient, the SLIM uses the beam divergence along the sample [20]. This section indicates how the divergence can be evaluated in a single-shot SLIM measurement.

It is possible to define effective parameters to characterize the beam width, curvature radius, and divergence angle using irradiance moments in real and spatial frequency spaces [28,29]. The propagation of these effective parameters can be solved exactly using the method of moments [30] under several scenarios, including quadratic media and NL refraction [31]. In summary, Ref. [31] shows in the paraxial regime how the effective beam width evolves according to the beam effective curvature, whose evolution in turn depends on the beam effective divergence angle,

$$\overline{\Theta^2} = \frac{\int d^2\mathbf{r} \mathcal{E}^* \left(\frac{-2\nabla_T^2}{k^2} \right) \mathcal{E}}{\int d^2\mathbf{r} |\mathcal{E}|^2}. \quad (18)$$

The parameter $\overline{\Theta^2}$ is conserved within a linear medium and is related with the beam quality factor M^2 [28,29]. A medium containing a pure refractive NL response $n' = n_2 I$ also has a conserved quantity involving $\overline{\Theta^2}$ at two z planes [31], which can be calculated with an extension of the reasoning outlined in the last section using the method of moments [30–32]. In order to generalize the results in these references, consider that

$$i \frac{d}{dz} \mathcal{E} = \hat{H} \mathcal{E} = (\hat{U} - i\hat{V}) \mathcal{E}, \quad (19)$$

$$i \frac{d}{dz} \mathcal{E}^* = -\hat{H}^* \mathcal{E}^* = -(\hat{U} + i\hat{V}) \mathcal{E}^*, \quad (20)$$

where $\hat{U}$ and $\hat{V}$ are real and linear operators that may contain the intensity, transverse position, or spatial derivatives. $\hat{U}$ and $\hat{V}$ act upon all terms to their right. Consider now some operator $\hat{O}$ that describes some wavefront quantity as the beam width, curvature radius, or divergence. $\hat{O}$ is similar to $\hat{U}$ and $\hat{V}$ and may involve transverse coordinates, spatial derivatives, or the intensity and acts upon all functions to its right as well. The average of $\hat{O}$ over the wavefront is defined as

$$\bar{O} = \frac{\int d^2\mathbf{r} \mathcal{E}^* \hat{O} \mathcal{E}}{\int d^2\mathbf{r} |\mathcal{E}|^2}. \quad (21)$$

The propagation of $\bar{O}$ can be evaluated through the z derivative of Eq. (21). The calculation of the term involving the derivative of the denominator in (21) is similar to that leading to Eq. (11), resulting in $\bar{O}2\bar{V}$. The z derivative of the

numerator's integrand requires some extra effort. Notice that

$$i \frac{d}{dz} (\mathcal{E}^* \hat{O} \mathcal{E}) = (-\hat{H}^* \mathcal{E}^*) \hat{O} \mathcal{E} + \mathcal{E}^* \left(i \frac{d\hat{O}}{dz} \right) \mathcal{E} + \mathcal{E}^* \hat{O} (\hat{H} \mathcal{E}) \quad (22)$$

resembles the evolution equation for Heisenberg picture operators in quantum mechanics [33]. To make this resemblance closer, it is assumed that the order of $-\hat{H}^* \mathcal{E}^*$ can be interchanged. This is a valid assumption for terms proportional to the intensity and spatial coordinates. Since $\hat{H}^*$ only involves second-order derivatives, two integration by parts are sufficient to rewrite (22) as

$$i \frac{d}{dz} (\mathcal{E}^* \hat{O} \mathcal{E}) = \mathcal{E}^* \left(\hat{O} \hat{H} - \hat{H}^* \hat{O} + i \frac{d\hat{O}}{dz} \right) \mathcal{E}. \quad (23)$$

In terms of $\hat{U}$ and $\hat{V}$, commutators (defined by $[\hat{a}, \hat{b}] = \hat{a}\hat{b} - \hat{b}\hat{a}$ for two operators $\hat{a}, \hat{b}$) and anticommutators (defined by $\{\hat{a}, \hat{b}\} = \hat{a}\hat{b} + \hat{b}\hat{a}$), Eq. (23) can be stated as

$$\frac{d}{dz} (\mathcal{E}^* \hat{O} \mathcal{E}) = \mathcal{E}^* \left(-i[\hat{O}, \hat{U}] - \{\hat{O}, \hat{V}\} + \frac{d\hat{O}}{dz} \right) \mathcal{E}. \quad (24)$$

The result in Eq. (24) is relevant since it suggests a conserved quantity in a nondissipative system. For $\hat{V} = 0$, dissipation is not present and an operator $\hat{O} = c \hat{U}$ for an arbitrary constant c also trivially satisfies $[c \hat{U}, \hat{U}] = 0$. Indeed, as will be discussed later, if $c = 4/k$, then the term proportional to the Laplacian becomes the effective divergence operator as defined in (18), which is conserved in linear optical systems [31]. The z derivative of $\bar{O}$ can now be written as

$$\frac{d\bar{O}}{dz} = \frac{\int d^2\mathbf{r} \mathcal{E}^* (-i[\hat{O}, \hat{U}] - \{\hat{O}, \hat{V} - \bar{V}\} + \frac{d\hat{O}}{dz}) \mathcal{E}}{\int d^2\mathbf{r} |\mathcal{E}|^2}, \quad (25)$$

where the term involving $\bar{V}$ comes from the denominator's derivative since $\hat{O}2\bar{V} = \{\hat{O}, \bar{V}\}$. The solution provided in (25) is generic and does not depend on any specifics of the medium and should be applicable in general. As a reminder, $\hat{O}$ can be associated with several effective beam quantities, as the beam width $\mathbf{r}^2$, or the curvature radius [31].

From now on, we will assume the specific model considered in this paper in Eq. (8) with two-photon absorption and third-order Kerr effect, where $\hat{U} = -\frac{1}{2k} \nabla_T^2 - k_0 n_2 I$ and $\hat{V} = \frac{\alpha_1}{2} + \frac{\alpha_2 I}{2}$. As previously stated, if $\hat{O} = \frac{4}{k} \hat{U} = \hat{O}^2 - 4 \frac{n_2}{n_0} I$, then the commutator term in (25) vanishes automatically, being a candidate for a conserved quantity when $\hat{V} = 0$ (no extinction). The choice $\hat{O} = \frac{4}{k} \hat{U}$ leads to the following term inside the numerator's integrand in Eq. (25), $\frac{d\hat{O}}{dz} = -4 \frac{n_2}{n_0} \frac{dI}{dz}$. Given the identity

$$\frac{d\bar{I}}{dz} = 2 \frac{d\bar{I}}{dz} + 2\bar{V}\bar{I}, \quad (26)$$

where it is remarked that the z derivative of I does not commute with the transverse averaging, it is possible to state that

$$\frac{d}{dz} \left(\frac{4}{k} \bar{U} + 2 \frac{n_2}{n_0} \bar{I} \right) = \frac{\int d^2\mathbf{r} \mathcal{E}^* \left(-\left\{ \frac{4}{k} \hat{U}, \hat{V} - \bar{V} \right\} + 4 \frac{n_2}{n_0} I \bar{V} \right) \mathcal{E}}{\int d^2\mathbf{r} |\mathcal{E}|^2}. \quad (27)$$

Notice that $K = \frac{4}{k} \bar{U} + 2 \frac{n_2}{n_0} \bar{I}$ is a quantity that is conserved when $\hat{V} = 0$. The last term inside the integrand comes from

identity (26). After an expansion of $\hat{U}$, $\hat{V}$ and rearrangement of the remaining terms in (27), it is possible to exactly verify that

$$\frac{dK}{dz} = -\alpha_2 h(I, z) + \frac{4n_2}{n_0} \left[\frac{\alpha_1}{2} \gamma P + \alpha_2 \left(\mu - \frac{\gamma^2}{2} \right) P^2 \right], \quad (28)$$

$$K = \overline{\Theta^2} - 2 \frac{n_2}{n_0} \gamma P, \quad (29)$$

$$\mu = \frac{\int d^2 \mathbf{r} I^3}{(\int d^2 \mathbf{r} I)^3}, \quad (30)$$

$$h(I) = \frac{\int d^2 \mathbf{r} \text{Re} \left[\mathcal{E}^* \left(\frac{-2\nabla_\perp^2}{k^2} \right) (I - \gamma P) \mathcal{E} \right]}{\int d^2 \mathbf{r} |\mathcal{E}|^2}. \quad (31)$$

In the absence of NL extinction, K is a conserved quantity and can be used to relate $\overline{\Theta^2}$ at two z planes [31]. Since P and γ are experimentally accessible parameters, the conservation of K can be used to obtain the divergence angle from the collected data in the limit $\alpha_1 \approx \alpha_2 = 0$. In the regime where $\alpha_2 \neq 0$, the right-hand side of (28) remains small in several practical scenarios. $h(I, z)$ is a transverse average of the divergence angle operator, $-2\nabla_\perp^2/k^2$, weighted by $(I - \gamma P)$ which averages to zero, and multiplied by the small quantity α_2 . Meanwhile, the second term in (28) contains products of the small quantities n_2 and α_i , where α_i should be relatively small in transparent materials where the SLIM can be applied. Therefore, K should be approximately conserved for sufficiently low powers and the beam divergence can be used in SLIM to evaluate the NL refraction, as previously performed in our group [23]. However, the present analysis is more complete and general than the one present in [23] and that in [31] since linear and NL extinction are included. In the D4 σ , the beam divergence is also an important experimental parameter to determine n_2 [15].

D. Transverse nonlinear light scattering

A scheme of the SLIM experiment is shown in Fig. 1. The camera collects light scattered at a 90° angle from the propagation direction. For simplicity, it is assumed that light is scattered because of the refractive index contrast between inclusions and the surrounding medium, Δn . Under such general conditions for a dilute medium, it may be stated in the Rayleigh-Gans approximation that the irradiance due to light scattered along the CCD direction can be written as

$$I_S(\mathbf{r}, z) = \eta_S f |\Delta n|^2 I_L(\mathbf{r}, z), \quad (32)$$

where the sample filling factor is f , $I_L(\mathbf{r}, z)$ is the beam irradiance at the inclusion location $(\mathbf{r}, z)$, and η_S is a proportionality factor associated with the scattering efficiency towards the camera direction. The local irradiance $I_L(\mathbf{r}, z)$ can include contributions due to forward- and backward-propagating light, but, for simplicity, we will only consider the forward terms and single scattering events. Since scattering will provide a power-loss mechanism for the propagating beam, it is possible to consider that

$$f |\Delta n|^2 \propto \alpha_{1,S} + \alpha_{2,S} I_L(\mathbf{r}, z), \quad (33)$$

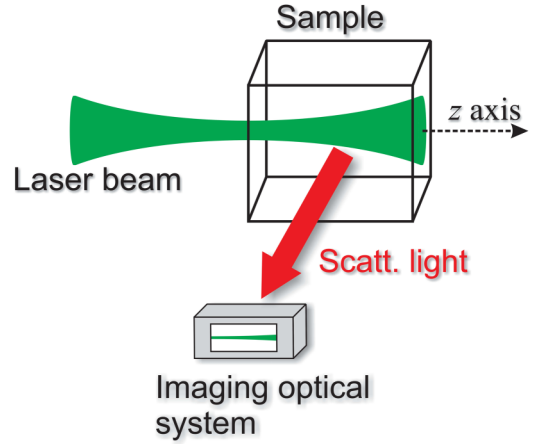


FIG. 1. Representation of the SLIM detection scheme. An optical beam propagating along the z axis is focused at the entrance of a scattering sample of length L . The light scattered along a transverse direction is collected by an imaging system containing a CCD camera. It is not strictly required to place the beam waist at the beginning of the sample, but this configuration is a good compromise between a strong NL response at the waist and still being able to measure the beam divergence when $L \approx 3z_0$, where z_0 is the Rayleigh length [20].

where $\alpha_{i,S}$ is related to the power extinction due to scattering. This work neglects NL scattering processes that generate new frequencies. The factor $\alpha_{2,S}$ indicates that NL scattering produces an observable effect, and it is central in the present discussion. Given that the SLIM is used to characterize NL samples through scattered light, it is crucial to understand the dependence of the collected signal due to NL scattering.

The SLIM camera detects light scattered transversely along the sample. Assuming that the scattered light is produced only by the forward-propagating field, the scattered light irradiance is given by

$$I_S(\mathbf{r}, z) = \eta_S \{ \alpha_{1,S} + \alpha_{2,S} I(\mathbf{r}, z) \} I(\mathbf{r}, z). \quad (34)$$

To consider the detected optical power as a function of z , $I_S(\mathbf{r}, z)$ is integrated along the transverse $\mathbf{r}$ plane. The optical power that reaches the camera, $P_C(z) = \eta_C \int d^2 \mathbf{r} I_S(\mathbf{r}, z)$, satisfies

$$P_C(z) = \eta [\alpha_{1,S} P(z) + \alpha_{2,S} \gamma(z) P^2(z)], \quad (35)$$

such that the detected light grows nonlinearly with the beam power. $\eta = \eta_C \eta_S$ indicates an overall detection efficiency of the system, while η_C is directly related to the camera detection efficiency.

If the NL scattering is negligible, then $\alpha_{2,S} = 0$ and Eq. (35) simplifies considerably to $P_C(z) = \eta \alpha_{1,S} P(z)$. Therefore, the power detected by the camera is proportional to the optical power inside the sample. Then, Eq. (16) may be used to evaluate how the beam extinction affects the optical power of a cw beam along the sample.

While this section considers corrections to the power measurements, similar corrections are necessary to relate all beam quantities at the sample to the camera measurements. The determination of $\alpha_{2,S}$, as discussed in the next section, allows one to correct the collected intensity profiles. Also, for NL

refraction measurements at sufficiently low powers, it should be reasonable to neglect the NL scattering corrections. The beam intensity is much smaller at the end of the sample, where the beam divergence angle is used to evaluate n_2 [20].

The presence of NL scattering requires extra steps to describe the detected optical power. Also, since several NL optics experiments use short pulses, it is important to consider the effects of the temporal averages over the pulse duration. The next section discusses how to describe the SLIM signal under such circumstances.

E. Temporal averaging

The temporal response is considered through a time averaging in which the NL response is considered to be instantaneous through the transformation $\bar{I}(z) \rightarrow \bar{I}(z)f(t)$, and the time averages are defined as $\langle O \rangle = \int dt O f(t) / \int dt f(t)$. From these definitions, the time average of Eq. (35), $\langle P_C(z) \rangle$, requires the calculation of $\langle P(z, t) \rangle$ and $\langle P^2(z, t) \rangle$, where $P(z, t)$ is the instantaneous optical power at a time t at the plane z . While in a real situation $\gamma(z)$ should be determined from the experiment, it is assumed for definiteness that $\gamma(z) = [\pi w_0^2(1 + \Delta z^2/z_0^2)]^{-1}$, where $\Delta z = z - z_c$, z_0 is the Rayleigh range and z_c represents the minimum beam-waist position in the linear regime. The time averaging does not depend on $\gamma(z)$, such that the following equations can be easily modified to consider other dependencies for $\gamma(z)$. Then, $P(z)$ given by Eq. (16) can be expressed as

$$P(z, t) = \frac{P_0 f(t) e^{-\alpha_1 z}}{1 + A(z) f(t)}, \quad (36)$$

$$A(z) = \alpha_2 I_0 \int_0^z \frac{e^{-\alpha_1 z'}}{1 + \Delta z'^2/z_0^2} dz'. \quad (37)$$

However, if it is true that $A(z)f(t) < 1$ for all z, t , it is then possible to use the geometric series expansion,

$$P(z, t) = P_0 e^{-\alpha_1 z} \sum_{i=0}^{\infty} [-A(z)]^i f(t)^{i+1}, \quad (38)$$

while on similar grounds,

$$P^2(z, t) = P_0^2 e^{-2\alpha_1 z} \sum_{i=0}^{\infty} (i+1) [-A(z)]^i f(t)^{i+2}. \quad (39)$$

The series expansions are useful for allowing the time averages to be taken only over the powers of $f(t)$. The temporal pulse profile is assumed to be Gaussian shaped, $f(t) = e^{-t^2/\tau^2}$. Then, it is possible to show that $\langle f(t)^k \rangle = k^{-1/2}$, and

$$\langle P(z, t) \rangle = P_0 e^{-\alpha_1 z} \Sigma_1[A(z)], \quad (40)$$

$$\langle P^2(z, t) \rangle = P_0^2 e^{-2\alpha_1 z} \Sigma_2[A(z)], \quad (41)$$

$$\Sigma_1[A(z)] = \sum_{i=0}^{\infty} \frac{[-A(z)]^i}{\sqrt{i+1}}, \quad (42)$$

$$\Sigma_2[A(z)] = \sum_{i=0}^{\infty} \frac{(i+1)}{\sqrt{i+2}} [-A(z)]^i. \quad (43)$$

It should be remarked that while the above expressions for $\Sigma_{1,2}$ are exact, both series quickly diverge in the limit $A \rightarrow 1$. As an alternative, it is possible to numerically evaluate $\Sigma_{1,2}$ or to use analytic approximations as those accurate to $\pm 1\%$ in the range $0 \leq A \leq 50$, as discussed in Appendix B.

With the previous assumptions, it can be stated that the time-averaged power detected by the camera is given by

$$\langle P_C(z, t) \rangle = \eta \alpha_{1,S} P_0 e^{-\alpha_1 z} \Sigma_1 \times \left[1 + \frac{\alpha_{2,S} I_0}{\alpha_{1,S}} \left(\frac{e^{-\alpha_1 z}}{1 + \Delta z^2/z_0^2} \right) \frac{\Sigma_2}{\Sigma_1} \right], \quad (44)$$

where Eq. (17) was used to obtain an expression in terms of I_0 . The logarithm of Eq. (17) leads to an expression which is linear in $\alpha_1 z$,

$$\ln \langle P_C(z, t) \rangle = C + \ln I_0 - \alpha_1 z + \ln \Sigma_1 + \ln \left[1 + \frac{\alpha_{2,S} I_0}{\alpha_{1,S}} \left(\frac{e^{-\alpha_1 z}}{1 + \Delta z^2/z_0^2} \right) \frac{\Sigma_2}{\Sigma_1} \right], \quad (45)$$

which can be very useful to understand in which range the NL response decreases its relevance. Equation (45) describes the detected optical power along a direction transverse to the beam propagation axis, and also includes effects due to NL scattering. If a given sample presents only two-photon absorption, $\alpha_{2,S} = 0$ and the logarithm involving Σ_2 becomes zero, and the analysis becomes quite simple. However, if $\alpha_{2,S} I_0 / \alpha_{1,S}$ is not negligible, then all terms in Eq. (45) are required to understand the detected optical power in the asymptotic limit $z \rightarrow L$.

III. CONSEQUENCES OF NL SCATTERING

This section discusses directly how NL scattering influences experimental measurements. An important remark is that NL-transmittance-based techniques (such as Z-scan, D4 σ , optical limiting, and others) measure $\langle P(z, t) \rangle$ at the end of the NL sample superimposed with the forward scattering. When the forward scattering is much smaller than the effective beam transmitted power, such techniques are mainly sensitive to the total extinction and cannot unambiguously distinguish NL absorption from NL scattering. The varying aperture approach [19,26] can be used to separate NL extinction from the forward-scattering contribution. On the other hand, the SLIM determines the NL contribution from the light scattered towards a camera in the transverse plane, so it may perform such distinction in principle. Extinction governs the power propagation, while the NL scattering contribution modifies the detected signal. This separability seems to be particularly important for composite media containing inclusions with a high refractive index contrast with respect to the surrounding medium. Colloids containing either metallic nanoparticles or transparent materials with large dielectric constants are strong candidates to exhibit large elastic NL scattering behavior.

For simplicity, we consider the SLIM signal due to a sample where the linear extinction is only due to scattering, $\alpha_1 = \alpha_{1,S}$, and that linear scattering is small across a sample of length $L = 3z_0$. The beam waist is positioned at the sample entrance, and $z_c = 0$. To parametrize the NL scattering, the

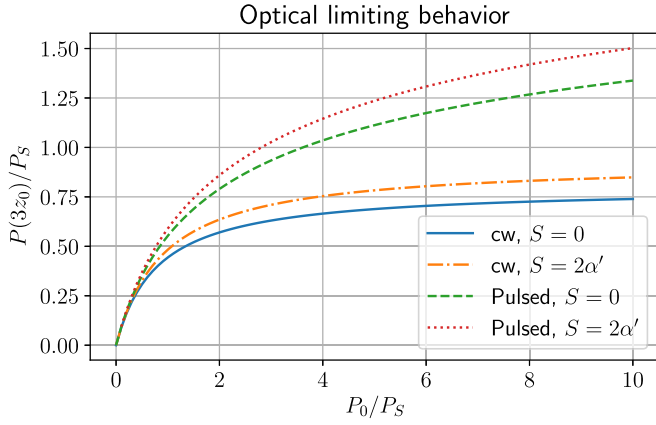


FIG. 2. Optical limiting at the end of the thick sample for pulsed (dashed and dotted lines) and cw (solid and dash-dotted lines) beams; $\alpha' = 10^{-3}$.

total extinction coefficient, $\alpha_2 = \alpha_{2,A} + \alpha_{2,S}$, is fixed and the parameter S is introduced such that $\alpha_{2,S} = S\alpha_2$ and $\alpha_{2,A} = (1 - S)\alpha_2$. S can be understood as the relative contribution of NL scattering to the NL extinction, while the total NL power extinction remains fixed. Only NL absorption is relevant when $S = 0$, while NL scattering is the only source of NL extinction for $S = 1$. Several normalization parameters are also introduced to express the calculations in terms of dimensionless parameters. The lengths along z are normalized by the Rayleigh length, z_0 . The total linear extinction coefficient becomes $\alpha'_1 = \alpha_1 z_0$. The optical power is normalized by $P_S = \pi w_0^2 / \alpha_2 z_0$, which is a scale factor related to the saturation of the transmitted power. Also, even though the approximation $\gamma(u) = [\pi w_0^2 (1 + u^2)]^{-1}$ necessarily breaks down upon large NL extinction, such refinements will not be considered here. In practice, $\gamma(u)$ should be measured directly. For pulsed beams, the time averages will be performed numerically, such that the following considerations are not limited to $P_0 \ll P_S$. The approximations in Appendix B could be used as well. Also, the power detected by the camera, $P_C(z)$, is normalized by its maximum linear detection efficiency, $\eta\alpha_{1,S}$.

Initially, it is important to understand the effects of the NL extinction over the measured signals, such that a small linear extinction is assumed, $\alpha' = 10^{-3}$. In Fig. 2 it is shown the optical limiting behavior associated with the power at the end of the sample under consideration. The tails of a Gaussian pulsed beam are subject to less NL extinction and more power gets transmitted. Meanwhile, the optical limiting behavior is more effective for a cw beam since the initial field amplitude is always at the average power. The qualitative aspects of pulsed and cw beams are identical in every aspect discussed below. Quantitatively, the cw case has an increased power extinction. Notice that Fig. 2 consists of the optical beam power, which is related to measurements in a typical NL transmittance experiment. Figure 2 also considers the effective optical limiting signal as would be perceived in SLIM for an arbitrary value $S/\alpha' = 2$. It can be perceived that both sets of curves (S/α' either 0 or 2) exhibit similar optical limiting behaviors, but larger values of S increase the detected optical power in SLIM. Therefore, the simple evaluation of

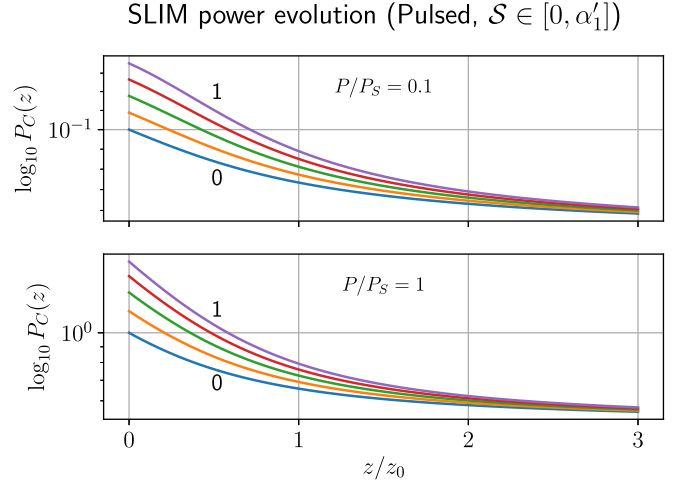


FIG. 3. Effects of an increased S on the SLIM power evolution curve for fixed extinction coefficients. S/α' increases sequentially in the displayed curves from the bottom to the top, with the extrema labeled for convenience. $\alpha' = 10^{-3}$ and $S/\alpha' = 0, 0.25, 0.5, 0.75, 1.0$.

the NL transmittance of a sample containing significant NL scattering can contaminate the measurement of α_2 , leading to incorrectly measured values in SLIM. This can also affect ordinary NL transmittance techniques, but in this case one must consider the contribution due to forward scattering along the sample and the collection optics aperture [17,19,26]. Since the D4 σ technique uses a camera to detect the transmitted beam, it is possible to vary the detection aperture digitally for each laser shot, instead of performing several measurements as in [19,26].

S can be determined directly through the evaluation of the power evolution profiles in SLIM. This approach retains one of the most relevant SLIM features, which is its ability to perform single-shot measurements. In Fig. 3, it is shown how the onset of NL scattering affects SLIM measurements. A larger value of S decreases the initial slope of $P_C(z)$, but retains the asymptotic behavior. Notice, therefore, that the SLIM allows one to distinguish the contributions of NL scattering and NL absorption to the total extinction. The asymptotic optical power does not strongly depend on S and can be used to determine α_2 . The SLIM power evolution signal at $z/z_0 < 1$ strongly depends on S , so it can be determined through a curve fitting of $P_C(z)$. This is a substantial aspect in favor of SLIM since typical NL transmittance techniques are only able to determine the total extinction coefficient. Given its single-shot capability [20], it becomes possible to consider a single-shot characterization of NL refraction, absorption, and scattering.

Other features of the power evolution curves as measured in SLIM are considered below. Figure 4 indicates the optical power evolution when the NL extinction is due to NL absorption only, $S = 0$, or when the NL extinction is due to NL scattering only, $S = 1$. An increased input power decreases the power at the output plane due to the normalization at the sample initial plane. The qualitative behavior of pulsed and cw beams is similar, but the smaller attenuation of pulsed beams leads to higher detected powers. The evolution profile is substantially modified from $S = 0$ to $S = 1$. Most of the

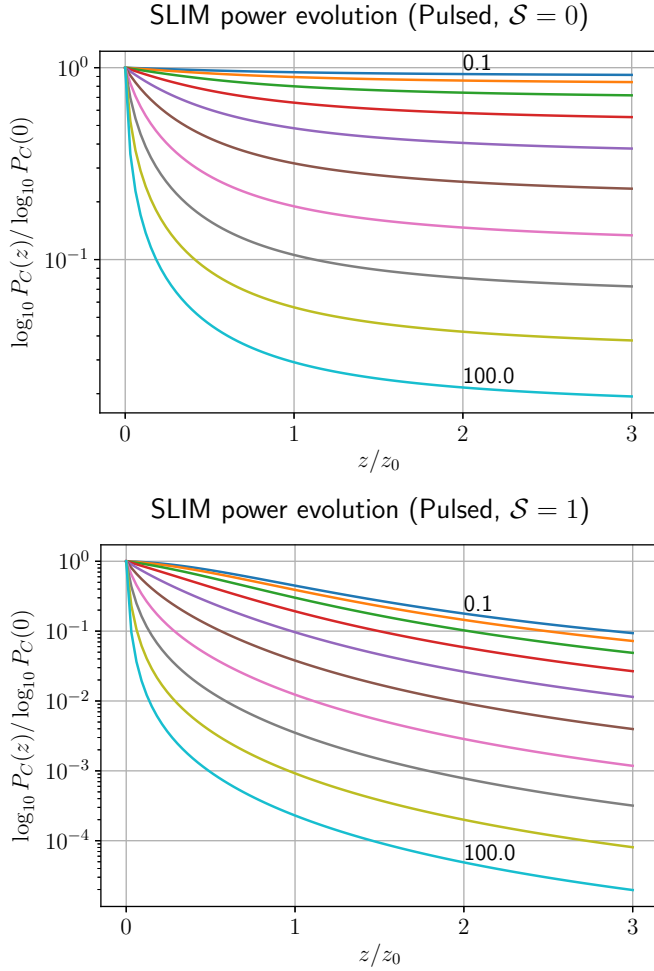


FIG. 4. SLIM power evolution signal when the two-photon extinction is only due to NL absorption ($S = 0$) or NL scattering ($S = 1$). The input optical power $P_0/P_s = 0.1, 0.2, 0.5, 1, 5, 10, 50, 100$, with the extrema labeled for convenience; $\alpha' = 10^{-3}$.

NL extinction occurs at the beginning of the sample, such that the SLIM signal decreases much faster upon propagation for $S = 1$. This characteristic indicates that an increased S demands more dynamic range from the detection system. Notice also that the asymptotic behavior changes substantially when S varies, having a steeper slope at $S = 1$.

It is also possible to verify how α' affects $P_C(z)$ for pulsed beams, as shown in Fig. 5. An increased linear extinction reduces the NL response, especially at small z/z_0 values. Notice that $P_C(z)$ increases significantly around $z \approx 0$ when the linear losses are sufficiently small. Larger α' values reduce both the z range, where the NL scattering is more prominent, and the signal dynamic range.

IV. SUMMARY

This work proposes in detail how to determine the NL optical constants of turbid media using the scattered light imaging method (SLIM) [20] together with a NL transmittance technique, such as Z-scan [13] or D4 σ [14,15]. The present proposal is important for filling a gap in the avail-

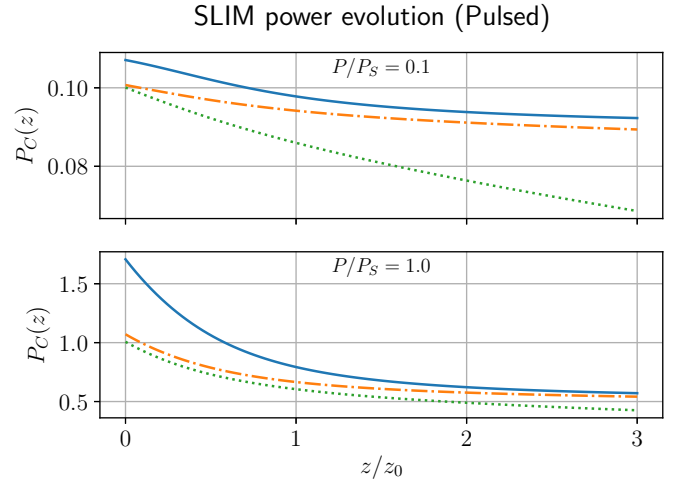


FIG. 5. SLIM power evolution signal dependence on the linear extinction coefficient α' . $S = 10^{-3}$ for both powers and α' assume the values 10^{-3} (solid line), 10^{-2} (dash-dotted line), and 10^{-1} (dotted line).

able methods to characterize the NL properties of several disordered matter systems. The analysis indicates that SLIM not only is very sensitive to NL scattering, since the signal increases significantly in the NL region, but it also can be used to distinguish NL absorption from NL scattering. For a fixed total power extinction, an increase in NL scattering substantially modifies the dynamic range and shape of the SLIM power evolution signal. Therefore, the total two-photon power extinction coefficient α_2 can be determined from a NL transmittance technique, and SLIM can be used to discriminate the ratio between NL scattering and NL absorption coefficients. These are important remarks from a spectroscopic point of view because NL transmittance characterizations cannot unequivocally distinguish power losses due to NL absorption from those due to NL scattering. Colloids containing metallic inclusions or with high refractive index contrast inclusions are excellent candidates to exhibit strong elastic NL scattering contributions, and the present considerations may imply corrections over some measurements obtained with NL transmittance techniques. Through the inclusion of NL absorption and NL scattering, this work extends the results of [23], where the NL refraction is measured using the SLIM. Since the SLIM and the spatially resolved NL transmittance in D4 σ can be determined from a single laser shot, the present discussion should be useful to understand the NL optical behavior of samples susceptible to optical damage, such as biological materials.

ACKNOWLEDGMENTS

This work was supported by Brazilian agencies Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq), Fundação de Amparo à Ciência e Tecnologia do Estado de Pernambuco (FACEPE), and Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES). The work was performed in the framework of the National Institute of Photonics (INCT de Fotônica) project and PRONEX/CNPq/FACEPE.

APPENDIX A: THE SOLUTION TO EQ. (15)

A nonlinear differential equation that can be written as

$$\frac{df(z)}{dz} = a_1 f(z) + a_2 g(z)[f(z)]^2 \quad (\text{A1})$$

can be solved, in general, for a generic known function $g(z)$ and constant coefficients a_0 and a_2 . The initial value problem $f(0) = f_0$ is considered in what follows. Instead of solving directly for $f(z)$, it is useful to perform the following functional transformation:

$$f(z) = \frac{f_0 e^{a_1 z}}{h(z)}, \quad (\text{A2})$$

where the condition $h(0) = 1$ is required to satisfy the initial value condition for $f(z)$. Then, the differential equation for $h(z)$ becomes

$$\frac{dh(z)}{dz} = -a_2 g(z) e^{a_1 z}. \quad (\text{A3})$$

Since $g(z)$ is known by hypothesis, Eq. (A3) can be directly integrated. Therefore, the full solution for $f(z)$ becomes

$$f(z) = \frac{f_0 e^{a_1 z}}{1 - a_2 \int_0^z g(z') e^{a_1 z'} dz'}. \quad (\text{A4})$$

APPENDIX B: ANALYTIC APPROXIMATIONS TO $\Sigma_{1,2}$

In this Appendix, analytic approximations to Σ_1 and Σ_2 are obtained. The expressions below should be accurate enough (better than $\pm 5\%$ and $\pm 1\%$) for most NL spectroscopic studies. The functions Σ_1 and Σ_2 are very important in the description of the SLIM experimental results, and the series expansions previously described in Eqs. (42) and (43) diverge whenever $A(z) \rightarrow 1$.

First, notice that both terms can be expressed as $\Sigma_i[A(z)]$, such the specific dependence on z comes only through $A(z)$. Therefore, it is sufficient to calculate an approximation to $\Sigma_i(A)$, where A is an independent variable. Given that

$$\Sigma_1(A) = \frac{\int_{-\infty}^{\infty} dt \frac{f(t)}{1+A f(t)}}{\int_{-\infty}^{\infty} dt f(t)}, \quad (\text{B1})$$

it is possible to see that in the limit of a cw beam, $\Sigma_1(A) = 1/(1+A)$. Meanwhile, if $f(t) = e^{-t^2/\tau^2}$, then the Laplace's approximation method to the above integral indicates that $\Sigma_1(A) \approx 1/(1+A)^{0.5}$. A numerical evaluation indicates that Laplace's approximation is useful to indicate the trend of $\Sigma_1(A)$. An educated-guess approximation is to least-squares optimize the exponent a in

$$\Sigma_1(A) = 1/(1+A)^a, \quad (\text{B2})$$

to better approximate the exact numerical solution. It can be found that $a = 73/94$ is accurate within $\pm 5\%$ for $0 \leq A \leq 30$, which significantly extends the range of intensities relative to the series expansion in Eq. (42). As a practical example, consider a colloid containing ZnSe inclusions at 532 nm and in the picosecond regime. It is known that bulk ZnSe has $\alpha_2 = 5.8 \text{ GW/cm}^2$ [13], such that this value should be an approximate upper bound for the colloid's total NL extinction when the solvent does not exhibit two-photon absorption. A negligible linear extinction allows one to estimate the

maximum value of A in the limit $z \rightarrow \infty$, or $A_{\max} = \alpha_2 I_0 z_0 \frac{\pi}{2}$. $z_0 = 0.3 \text{ cm}$ is a practical value for a liquid sample since the total sample length must be $3z_0$. Given that the series expansion is accurate for $A < 1$, the series solution is valid for irradiances $I_0 < 0.37 \text{ GW/cm}^2$. In the Z-scan technique, the NL transmittance varies 10% with $I_0 = 0.21 \text{ GW/cm}^2$. Since, at least for NL refraction, both the SLIM and D4 σ techniques reported an enhanced sensitivity with respect to Z-scan measurements [15,23], the series approximation should already be sufficient to obtain a good description of the NL extinction signal. In particular, considering Eq. (16) for $\alpha_1 \approx 0$, the condition $A = 1$ indicates that approximately half of the beam power is lost due to the NL response, such that the SLIM technique should be able to easily determine such extinction. Therefore, the best approximation in Eq. (B2) is accurate within $\pm 5\%$ for $I_0 < 11 \text{ GW/cm}^2$, significantly extending the range of intensities that can be experimentally analyzed.

The inclusion of a scale factor on A improves the approximation even more, and it can be stated that for Gaussian-shaped pulses,

$$\Sigma_1(A) = \frac{1}{(1+bA)^a}. \quad (\text{B3})$$

Using $a = 27/32$ and $b = 47/59$, $\Sigma(A)$ is accurate within $\pm 1\%$ for $0 \leq A \leq 50$, and accurate within $\pm 5\%$ for $0 \leq A \leq 100$. Therefore, Eq. (B3) can be considered as exact for most experimental and data-fitting purposes. Returning to the ZnSe colloid example, Eq. (B3) is accurate within $\pm 5\%$ for $I_0 < 37 \text{ GW/cm}^2$ when $\alpha_2 = 5.8 \text{ GW/cm}^2$. Smaller α_2 values extend the valid range of intensities, such that a colloid which has $\alpha_2 = 2 \text{ GW/cm}^2$ can be described up to $I_0 < 100 \text{ GW/cm}^2$. Higher-order NL effects may become evident in the SLIM signal at such high irradiances, and therefore we expect that the present model containing only α_2 loses its validity.

A similar analysis can be performed for

$$\Sigma_2(A) = \frac{\int_{-\infty}^{\infty} dt \frac{f(t)^2}{[1+A f(t)]^2}}{\int_{-\infty}^{\infty} dt f(t)}. \quad (\text{B4})$$

The Laplace's approximation suggests solving (B4) as $1/\sqrt{2}(1+A)^a$, with $a = 1.5$. The value $a = 152/91$ minimizes the least-squares distance between the educated guess and the full numerical solution within $\pm 5\%$ in the range $0 \leq A \leq 7$. The $\pm 1\%$ convergence range for $\Sigma_2(A)$ cannot be obtained as easily over a wide range of input powers as before over the same range of A . However, assuming a polynomial expansion over A as

$$\Sigma_2(A) = \frac{1}{\sqrt{2}} \frac{1}{(1+bA+cA^2+dA^3)^a}, \quad (\text{B5})$$

it can be found that the set of parameters $a = 191/320$, $b = 257/94$, $c = 129/53$, and $d = 58/89$ is accurate within $\pm 1\%$ for $0 \leq A \leq 70$ and remains within $\pm 2\%$ for $0 \leq A \leq 100$. Therefore, for most practical purposes, Eqs. (B3) and (B5) should be accurate to describe the propagation of Gaussian pulses in SLIM.

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